MIL-I-83262B(USAF)
25 May 1977
SUPERSEDING
MIL-I-83262A(USAF)
19 September 1972

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, HERMETICALLY SEALED, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hermetically sealed temperature indicators of the thermocouple-circuit type.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic

MILITARY

MIL-P-116	Preservation-Packaging Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-W-5846	Wire, Electrical, Chromel and Alumel, Thermocouple
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-S-20708	Synchros, General Specification for
MIL-B-22191	Barrier Materials, Transparent, Flexible, Heat Sealable

FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White General Specification for
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover
MIL-I-83262/1	Indicator, Temperature, Thermocouple EHU-31A/A
MIL-I-83262/2	Indicator, Temperature, Thermocouple EHU-38/A
MIL-I-83262/3	Indicator, Temperature, Thermocouple, EHU-43/A

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Circular 561	Reference Tables for Thermocouples

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between requirements of this specification and the detail specification, the latter shall govern.

3.2 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Preliminary list of electronic parts. A preliminary list of electronic parts shall be furnished directly to the procuring activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the contract quantity. The parts shall conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to fabrication of the complete item. Standard electronic parts shall be used wherever practicable. All discrete capacitors and fixed resistors, except temperature compensating resistors, shall be standard parts. Thermocouple wire shall be in accordance with MIL-W-5846.

5 Materials

3.5.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.5.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.5.3 Metals. Metals shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.5.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.4 Toxic and corrosive fumes. The materials, as installed and under the service conditions specified herein, shall not liberate gases or fumes that will result in corrosion to such extent as to cause malfunction of the indicator or discoloration of dials and indicia or be detrimental to the performance of the aircraft or to the health of personnel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.5.5 Protective treatment. When materials are used in the construction of the indicators that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5.6 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. External materials which are nutrients for fungi shall be treated with a fungicidal agent approved by the procuring activity.

3.6 Design and construction. The indicator shall be designed to indicate temperature when used in conjunction with a chromel-alumel thermocouple circuit with thermocouple signals in accordance with National Bureau of Standards Circular 561. It shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Major components and other design criteria shall be specified in the detail specification.

3.6.1 Power supply. The indicator shall be designed to operate within the tolerances specified in the detail specification when supplied with the following electrical power. Synchro devices within the indicator shall be in accordance with MIL-S-20708 and shall operate from the same phase of the power source that supplies power to the indicator. Where any instrument requires power other than normal 115V 400 Hz or 28V dc, it shall be generated within the electronic module itself. Internal lighting shall receive power from a 0-5V ac or dc source external to the indicator.

3.6.1.1 DC Powered indicators. The indicator shall be powered with 28V dc electrical power conforming to MIL-STD-704A, equipment Category B, for normal power conditions (NSSL), abnormal power conditions (ASSL), and emergency power conditions (ESSL).

3.6.1.2 AC Powered indicators. The indicator shall be powered with 115V 400 Hz ac electrical power conforming to MIL-STD-704A, equipment Category B, except that the voltage may vary ±20 percent and the frequency may vary ±20 percent. Operation with voltage surges to 195 ±5V shall be as specified herein and in the detail specification.

3.6.2 Power consumption. The indicator steady-state power consumption shall not exceed tolerance set in detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.6.3 Reliability. When reliability is required, it shall be in accordance with MIL-STD-781 and as specified in the detail specification.

3.6.4 Maintainability. The indicator shall be so designed as to provide ease of maintenance. The following requirements shall be included in the design of the indicator:

a. Rapid and positive identification of replaceable defective parts, assemblies, or components.

b. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance.

c. Optimum accessibility and replaceability of all components requiring maintenance.

3.7 Performance. The indicator shall meet all the requirements of the tests specified in Section 4 and in the detail specification.

3.7.1 Electromagnetic interference. The indicator shall meet the electromagnetic interference and susceptibility requirements of MIL-STD-461 equipment Class A1.

3.7.2 Magnetic effect. When positioned 5 ±.50 inches from a precision magnetic compass and in a terrestrial magnetic field intensity of 0.17 to 0.19 oersted at the indicator, compass deflection shall not exceed 5°.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Case. The case dimensions shall be in accordance with the detail specification. The case shall be hermetically sealed and so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools or fixtures unless they are approved by the procuring activity. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that would be affected by any atmospheric conditions to which the indicator might be subjected.

3.9.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either a mixture of 88 to 92 percent nitrogen and the remainder helium or 100 percent nitrogen. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

3.9.2 Filler tube. A filler tube or other means for evacuating and filling the case shall be provided on the rear of the case. If a filler tube is used, it shall be recessed beneath the back plate surface.

3.9.3 Nameplate. A nameplate shall be provided in the rear of the case. Metal decals shall not be used. The information on the nameplate shall conform to MIL-STD-130. The date of manufacture in terms of month and year shall be included on the nameplate of each indicator and shall be identified as such.

3.9.4 Case finish. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, Color No. 37038 of FED-STD-595. The remainder of the case shall be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein.

3.10 Dial. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not turn or loosen under vibration. The dial blank shall be a flat disc of metal, plastic, or other material that will permit compliance with all requirements of this specification.

3.10.1 Dial marking. The dial shall be marked as specified in the detail specification. The form of the numerals and letters shall be in accordance with MS33558. Lighted dials shall be marked in lusterless white, Color No. 37875 of FED-STD-595.

3.10.2 Dial visibility. The pointer, numerals, at least .03125 inch of the shortest graduations and all other specified markings on the dial shall be visible from any position in front of the dial ranging between normal and a position 30° to the normal.

3.11 Lighting. Internal lighting shall be designed into the indicator. The design of the indicator shall be such that the internal lighting may be either white or red depending upon the color specified. With the exceptions specified herein, the red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. The internal lighting of the prime standard and all qualification indicators shall be subject to the approval of the qualifying activity.

3.11.1 Lamps. Lamps shall be in accordance with MIL-L-25467 or MIL-L-27160.

3.11.2 Prime standard. The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an actual indicator conforming to the requirements of this specification or a lighted model. The prime standard shall be used as the reference for acceptance of all production indicators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.11.2.1 Prime standard recertification. All representative areas of the face of the prime standard indicator shall be measured for brightness in foot-lamberts. Each area shall be measured at a sufficient number of locations to obtain data representative of each area including the brightest and dimmest portions of the indicator in each display. The prime standard indicator shall be measured periodically (at least every 50 hours of operation or every 120 days, whichever occurs first) to determine any brightness or color change. The data and, if required by the qualifying activity, the prime standard shall be submitted for evaluation and reapproval or corrective action, or both.

3.11.2.2 Visual comparison. The lighting level of the indicator shall fall within the limits of the prime standard when energized at the voltage levels specified in 4.6.36.2.2.

3.12 Dials and pointers. Integrally lighted dials and pointers shall conform to MIL-L-25467 or MIL-L-27160, as specified in the detail specification. Trans-illuminated dials and pointers may be made of nonmetallic materials, and floodlighted dials and pointers shall be made of metal.

3.12.1 Pointer. The design of the pointer shall be as specified in the detail specification. The pointer shall be as light as practicable and sufficiently rigid to prevent flexing under vibration. Unless otherwise specified, the minimum length of the portion of the pointer that is visible shall be 0.50 inch. Unlighted pointers shall be channeled to provide adequate strength. Lighted pointers shall be lusterless white, Color No. 37875 of FED-STD-595.

3.12.2 Overswing. Pointer stops shall be provided and so located as to permit the pointer to overswing as specified in the detail specification.

3.12.3 Parallax. To minimize parallax, the clearance between the pointer and the graduated scale shall not exceed 0.10 inch.

3.12.4 Pointer marking. The pointer shall be marked as specified in the detail specification.

3.13 Cover glass. The cover glass shall be clear, flat and free from flaws which would interfere with the normal reading of the instrument. The cover glass shall be suitable for hermetic sealing and shall comply with the requirements of MIL-G-81704 for type III glass. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-14806.

3.13.1 Distance from dial to cover glass. The distance between the inside of the cover glass and the dial shall not exceed 0.188 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.14 Temperature signals. The indicator shall be designed to operate on signals generated by chromel-alumel thermocouples. The values of the signals shall be as specified in National Bureau of Standards Circular 561.

3.15 Grounding of thermocouple extension wires. The indicator shall be so designed that when either the chromel or alumel conductor is grounded one at a time, the temperature indication will not change by more than 2°C. The ground may be at any point between the thermocouple and the interior of the indicator.

3.16 Open thermocouple circuit or power failure. An open thermocouple circuit or an input power failure shall cause the indicator pointer to drive down-scale to or beyond the low end of the scale at a minimum average rate of 40°C per second from any point on the scale.

3.17 External thermocouple circuit resistance. When the thermocouple circuit resistance is varied from 5 to 200 ohms, the temperature indication shall not change by more than 2°C.

3.18 Mask. When required by the detail specification, a thin metal mask shall be provided directly behind the cover glass to cover all moving parts of the mechanism except the pointer and to provide for a visible uniform length of pointer regardless of the pointer indication. The mask shall be finished in durable lusterless black, Color No. 37038 of FED-STD-595 and shall be constructed in accordance with the detail specification.

3.19 Balance. When balance is required the moving element of the indicator shall be balanced by an approved means (such as threaded lock or split nuts or a wire helix that moves along a crossarm and counterweight arm) with an easily accessible means for rebalancing. Solder, shellac, or similar means for holding the balance weight shall not be used.

3.20 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.21 Internal wiring. Where practicable, internal wiring shall be tinned flexible stranded wire.

3.22 Thermocouple cold junction compensation. The indicator shall incorporate temperature compensation for the thermocouple cold junction within the indicator. Temperature control at the cold junction shall not be used. The compensation shall be such that the input temperature signal to the amplifier circuit will be independent of the temperature changes at the cold junction within the accuracy requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

3.23 Zero adjustment. When required by detail specification, the indicator shall be provided with a zero adjustment screw which shall be accessible externally from the rear of the case. The zero adjustment shall have sufficient friction that it will not change its position due to vibration encountered in service.

3.24 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.25 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.26 Conductive plastic potentiometers. Nonmetallic materials used in construction of potentiometers shall not absorb a quantity of moisture greater than 0.2 percent by weight when exposed to an environment of 95 percent humidity at 49°C for a period of 14 days. Additionally, conductive plastic potentiometers must be approved by the procuring activity prior to design of the indicator. The Government reserves the right to both require contractor tests and conduct Air Force laboratory tests on conductive plastic potentiometers proposed for use.

3.27 Numerals and letters. Numerals and letters shall conform to MS33558.

3.28 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.29 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.29.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.29.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw or bolt or threads.

3.29.3 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. When required for accurate adjustments, gear assemblies shall be free from backlash.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.29.4 Cleaning. Care should be taken throughout various stages of assembly to thoroughly clear off loose, spattered, or excess solder, metal chips, and other foreign material. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be as specified herein and in the detail specification. The detail specification shall take precedence. The inspection and testing of indicators should be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Power supply. The power input to the indicator shall be as specified in the detail specification. Unless otherwise specified, the voltage and frequency shall be maintained with ±1 percent.

4.3.4 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken. If vibrated, the vibrator shall have a frequency within the 30- to 60-Hz range and an amplitude of 0.0015 ±0.0005 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.3.5 Mounting of instruments for tests. Unless otherwise specified, the instrument under test shall be mounted properly oriented and supported in a suitable metal panel in the same manner that it normally will be mounted and supported in the aircraft.

4.3.6 Test sequence. All tests shall be conducted in the order specified herein.

4.4 Qualification testing (see 6.3)

4.4.1 Test samples. The test samples shall consist of eleven indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Eight of the samples shall be subjected to the reliability tests only, and the other three shall be subjected to all the tests except reliability.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product (4.6.1)

b. Burn-in (4.6.2)

c. Electrical zero when applicable (4.6.3)

d. Scale error and friction (4.6.4)

e. Position error (4.6.5)

f. Warmup (4.6.6)

g. Response time (4.6.7)

h. Sealing (4.6.8)

i. Individual tests of MIL-L-27160 or MIL-L-25467 as applicable.

4.5.2 Sampling tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.5.2.1 Sampling plan A. Three indicators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6 and in the applicable lighting specification. For small contracts, those of 50 units or less, the following tests shall be performed on three indicators selected at random from each 50 produced since sampling plan A tests were last performed. If some quantity of the indicators has not been produced within 4 months of the present contract or order, sampling plan A tests shall be conducted on at least one of the units produced on the present contract. In the event several contracts or orders are filled simultaneously, three indicators from each 100 of the total quantity shall be subjected to the sampling plan A tests and the test results shall apply to the total quantity.

a. Individual tests (4.5.1)

b. Vibration (4.6.14)

c. Pointer oscillation (4.6.14.1)

d. Pointer variation (4.6.14.2)

e. Low temperature compensation (4.6.11)

f. High temperature compensation (4.6.12)

g. Resistance (4.6.17)

h. Magnetic effect (4.6.21)

i. Fogging (4.6.29)

j. Sampling plan A tests of MIL-L-27160 or MIL-L-25467, as applicable (integrally lighted indicators only)

All indicators subjected to sampling plan A testing may not be delivered on contract unless the indicator is completely refurbished (see 6.3.4) after sampling plan A tests.

4.5.2.2 Sampling plan B. Three indicators shall be selected at random from the first 15 or less produced on the contract or order and subjected to the following tests as described under 4.6 and in the applicable lighting specification. Unless so specified by the procuring activity (see 6.2), those sources listed on the applicable Qualified Products List will not be required to conduct these tests.

a. Sampling plan A (4.5.2.1)

b. Shock (4.6.10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

c. Low temperature exposure (4.6.15)

d. High temperature exposure (4.6.16)

e. Thermal shock (4.6.28)

f. Salt fog (4.6.30)

g. Humidity (4.6.31)

h. Detail examination (4.6.37)

i. Sampling plan B tests of MIL-L-27160 or MIL-L-25467, as applicable (integrally lighted indicators only).

4.5.2.3 Sampling plan C. When specified (see 6.2d), MIL-STD-105 sampling and inspection procedures shall be applied for Government acceptance of all lots or production indicators on contract. General Inspection Level II of table I, MIL-STD-105 shall be used throughout the acceptance program. Samples drawn for inspection shall be randomly selected by the Government representative after the whole lot has been submitted for acceptance. The acceptable quality level (AQL) to be maintained is 2.5. The sampling plan specified in table II-A, MIL-STD-105 shall be used initially. The reduced sampling plan of table II-C may be used when the criteria for reduced sampling has been attained and with approval of the procuring activity. A tightened inspection shall not apply without approval of the procuring activity. Each sample selected shall be subjected to and shall meet all tests listed under 4.5.1 prior to acceptance of the lot by the Government.

4.5.3 Defective units. A defective unit is an indicator which has failed any test listed under individual test requirements of this specification. The unit shall be considered defective as soon as it has failed a test requirement. A unit which has failed any test at only one test point shall be reinspected by the Government at least one time prior to being classified as defective.

4.5.3.1 Defective lots. Defective lots will be rejected and returned to the contractor. The lot shall not be resubmitted to the Government for acceptance until, as a minimum, it has been completely reinspected by the contractor. Repairs shall be made as appropriate prior to complete retesting by the contractor. Samples that are subjected to and pass all individual tests shall not be returned to the contractor with the remainder of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

the defective lot but shall be accepted as a production item. When resubmitted to the Government for acceptance, the defective lot shall again be subjected to the same acceptance criteria as specified in 4.5.2.3.

4.5.4 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.4.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.4.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, dimensions, markings, and workmanship. Prior to assembly, dials and pointers shall be inspected as to markings, dimensions, and finishes.

4.6.2 Burn-in. The indicator shall be placed in a chamber and the temperature maintained at 50°C for a minimum of 24 hours. Power shall be applied in accordance with 4.3.3 throughout the test while a constant signal input, which is determined at the beginning of test, is maintained to provide an indication of 750°C. At least 8 hours of this test shall be conducted during normal working hours, and observations shall be made and recorded at least once each hour throughout the 8-hour period. The indication change throughout the test shall not exceed 5°C. Pointer oscillation or erratic movement of the pointer shall be cause for rejection. The indicator shall be returned to room temperature for a minimum period of 2 hours after which it shall meet the test specified in 4.6.4.

4.6.3 Electrical zero (when applicable). For the purpose of this specification the electrical zero test and verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 115 ±2V 400 ±5Hz single-phase shall be applied to the rotor of the synchro with leads designated R1 and R2. R2 shall be connected to the ground or low potential side of the power input. With a lead designated S2 connected to R2, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S1 and S3 shall be minimum (usually 20 to 50 millivolts)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

b. The voltage between R1 and S3 shall be near the low null (approximately 104.8V) and shall reach the low null (approximately 103.2V) when the rotor is rotated clockwise approximately 30° from electrical zero.

c. Clockwise rotation of the rotor not to exceed 30° from electrical zero shall cause an increase in the voltage between S3 and S2 and a decrease in the voltage between S2 and S1.

4.6.4 Scale error and friction. The millivoltage values specified in National Bureau of Standards Circular 561 for chromel-alumel thermocouples, 0°C reference for the test points specified in the detail specification, shall be applied to the indicator. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in the detail specification. Upon completion of or concurrent with the scale error test, the friction error shall be determined by slowly changing the millivolt input between test values until the next millivolt test value is reached. When a test value is reached, the pointer indication shall be recorded before and after the indicator is lightly tapped or vibrated. The difference between the first and second reading at each test point shall be recorded as friction error. The difference between the second reading and the specified test point shall be recorded as scale error when these tests are conducted concurrently. The friction error test shall be conducted up-scale and down-scale at points pecified in the detail specification. The indicator shall meet the friction error tolerances specified in the detail specification. For indicators incorporating synchros, during this and subsequent scale error tests, the synchro output shall be within ±2°C of the indicated temperature, for temperatures within the expanded portion of the dial.

4.6.5 Position error. This test shall be conducted at the test points specified in the detail specification with power applied while the indicator is rotated 180° clockwise and 180° counterclockwise, held with the dial in the horizontal plane, face up and face down. With the millivoltage constant within ±0.02 millivolt, the position error in any of the specified positions shall not exceed the tolerances specified in the detail specification.

4.6.6 Warmup. Before power is applied, millivoltage shall be adjusted and applied to provide an input signal equivalent to 1,000°C. Power shall then be applied and the time required for the pointer to reach a stable indication, within ±15°C of the 1,000°C indication, shall be noted and shall not exceed 3 seconds. A minimum of 1 hour shall elapse between this power application and any previous power application. During the 1-hour period, no power shall be applied and the indicator shall be maintained at ambient test temperature.

4.6.7 Response time. Response time is to be tested in accordance with the detail specification tests and tolerances

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.8 Sealing. The indicator case shall be tested for leaks by means of a helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi. As an alternate the indicator shall be immersed in a suitable liquid, such as water. The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch Hg and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is longer. The absolute pressure shall then be increased to 2.50 inches Hg. Any bubbles coming from within the indicator case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as a leak.

4.6.9 Sensitivity. Millivoltage and power shall be supplied to provide an indication of 750°C. A change in the millivoltage equivalent to 2°C shall be made and a change of 2° ±1°C in indication shall result. This procedure shall be repeated at test points specified in the detail specification. The results shall be as specified for the 750°C point.

4.6.10 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.2, procedure I, figure 516.2-1, parameters (a) and (c), except that the acceleration shall be 10g in lieu of 20g. One shock in each direction shall be applied with no power on the indicator or its lighting circuit. During all other shocks, input power, including power for integral lighting, and input signal to produce an indication of 700°C shall be applied. No damage to the indicator which can be observed through visual inspection or can be determined through subsequent tests shall result.

4.6.11 Low temperature compensation. The indicator shall be placed in a temperature chamber at room temperature. Power and millivoltage shall be applied to produce a steady indication of 750°C. While maintaining the millivoltage constant within 0.02 millivolt, the temperature of the chamber shall be reduced to -40° ±2°C at a rate of approximately 5°C per minute and held at this temperature for 4 hours. While the temperature is being reduced, the indication shall not change by more than 5°C. Upon reaching the -40°C temperature, the indication shall be reduced to 100°C and the power shall be removed until the end of the 4 hours. The millivolt input shall then be adjusted to a value corresponding to 1,000°C, power shall be reapplied, and the warmup, response time, sensitivity, and scale error tests shall be conducted. The warmup time shall not exceed 6 seconds, the response time shall not exceed 2.5 seconds, and the sensitivity shall be as specified in 4.6.9. The scale errors shall not exceed the tolerances specified in the detail specification. During warmup the time shall be measured for the pointer to come within 15°C of 1,000°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.12 High temperature compensation. The indicator shall be tested as specified in 4.6.11, except that the temperature of the chamber shall be raised to 100°C. The warmup time, response time, and sensitivity shall be as specified therein and the scale errors shall not exceed the tolerances specified in the detail specification.

4.6.13 Effect of clutch and stops. The indicator shall be operated to indicate a temperature of 200°C. The input millivolts shall be instantaneously increased to 52 millivolts, maintained for 30 minutes, and reduced to provide an indication of 1,000°C. The millivoltage shall then be instantaneously reduced to -3 millivolts, held for 30 minutes, and increased to provide an indication of 200°C. This shall constitute one cycle and shall be repeated until eight cycles have been completed. The scale error test specified in 4.6.4 shall then be repeated and the results shall be as specified in table I or II, as applicable, for scale error at room temperature. A zero reference junction temperature shall be maintained throughout the test. This test shall be conducted with the indicator in an ambient temperature of 50° ±2°C.

4.6.14 Vibration. The indicator shall be subjected to vibration as specified by the detail specification. Pointer or counter oscillation shall not exceed 5°C either side of test point reference. The test point shall be 750°C with power applied. Following the vibration test, the indicator shall pass the test specified in 4.6.4. No screws or other parts shall become loosened or damaged as a result of this test. The equipment shall be considered hard mounted in service. Variation of indication from the test point at any time during the test shall be less than 5°C.

4.6.15 Low temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of -65° ±2°C for a period of 24 hours without power applied. While still at this temperature, the indicator shall be operated throughout the scale range without exhibiting erratic characteristics. The temperature shall then be raised to -40° ±2°C and maintained for a period of 24 hours without power applied. While still at this temperature, power shall be applied to the indicator and the warmup test and response test shall be conducted immediately. The warmup time shall not exceed 6 seconds and the response time shall not exceed 2.5 seconds. The indicator shall then be subjected to the scale error test at low temperature. The scale errors shall not exceed the tolerances specified in the detail specification for low temperature.

4.6.16 High temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of 100°C for a period of 24 hours. The indicator shall be powered to give an indication of 700°C throughout the last 2 hours of this period. While still at high temperature, the indicator shall be subjected to the scale error test. The scale error shall not exceed the tolerances specified in the detail specification for high

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

temperature. The indicator shall be returned to room temperature for a minimum period of 2 hours and again subjected to the scale error test. The scale errors shall not exceed the tolerances specified in the detail specification for room temperature. The indicator shall then be subjected to the response time test. The response time shall not exceed 2 seconds.

4.6.17 Thermocouple lead resistance effect. When the thermocouple lead resistance is varied between 5 and 200 ohms, the change in indication shall not exceed 2°C. The test shall be conducted at the test points specified in the detail specification.

4.6.18 Grounding of thermocouple extension wires. The chromel (+ conductor) and the alumel (- conductor) shall be alternately connected to the ground or low potential side of the input power to the indicator for a period of 1 minute. Prior to the grounding of the wire, the scale error and the response time shall be determined on increasing temperature at test points specified in the detail specification. The scale error and response tests shall be repeated with each lead grounded alternately. The scale error with either lead grounded shall not differ from the scale error without a lead grounded by more than 2°C. The response time with a lead grounded shall not differ from the response time without a lead grounded by more than 1 second.

4.6.19 Temperature-altitude. The indicator shall be subjected to an absolute pressure equivalent to an altitude of approximately 100,000 feet for a period of 1 hour at a temperature of -40° ±2°C. While still at this pressure and temperature, the indicator shall be subjected to the scale error test. The scale errors shall not exceed the tolerances specified in the detail specification for scale error at low temperature. There shall be no damage to the indicator as a result of this test.

4.6.20 Electromagnetic interference. The indicator shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, test methods CE01, CE02, CE03, CE04, CS01, CS02, CS06, RE02, RS01, RS02 and RS03.

4.6.21 Magnetic effect. The indicator, first not operating and then operating on rated voltage, shall be rotated in a vertical plane about a short-bar precision magnetic compass with the nearest part of the indicator 5 ±0.50 inches from and magnetically East or West of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The terrestrial horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 5°. The test point shall be 750°C during this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.22 Voltage surge, DC indicators. The indicator shall be supplied with 28V dc power and subjected to a series of five voltage surges which shall conform to the transient surge dc voltage step function limits of MIL-STD-704, figure entitled, Transient surge dc voltage step function loci limits for category B equipment, curve 1. At the conclusion of this test, the scale error shall not exceed the tolerances specified in the detail specification for room temperature. The test point shall be 750°C during this test.

4.6.23 Voltage variation, DC indicators. The indicator shall be subjected to the scale error test at room temperature first with the indicator receiving 16V dc mean power with a ripple as specified by MIL-STD-704A at 20 Hz and then with the indicator receiving 30V dc mean power with the same ripple. The indicator shall not exhibit any fluctuation or erratic conditions during this test. The scale error tolerances shall be 15 percent greater than the tolerances for scale error at room temperature.

4.6.24 Voltage and frequency variation, AC indicators. The indicator shall be subjected to the scale error test specified in 4.6.4 at room temperature. The power supply shall be varied to provide any combination of voltage and frequency ranging from 95 to 135V and 320 to 480 Hz during each phase of the test. The change in indication at the varied voltages and frequencies shall not exceed 5°C.

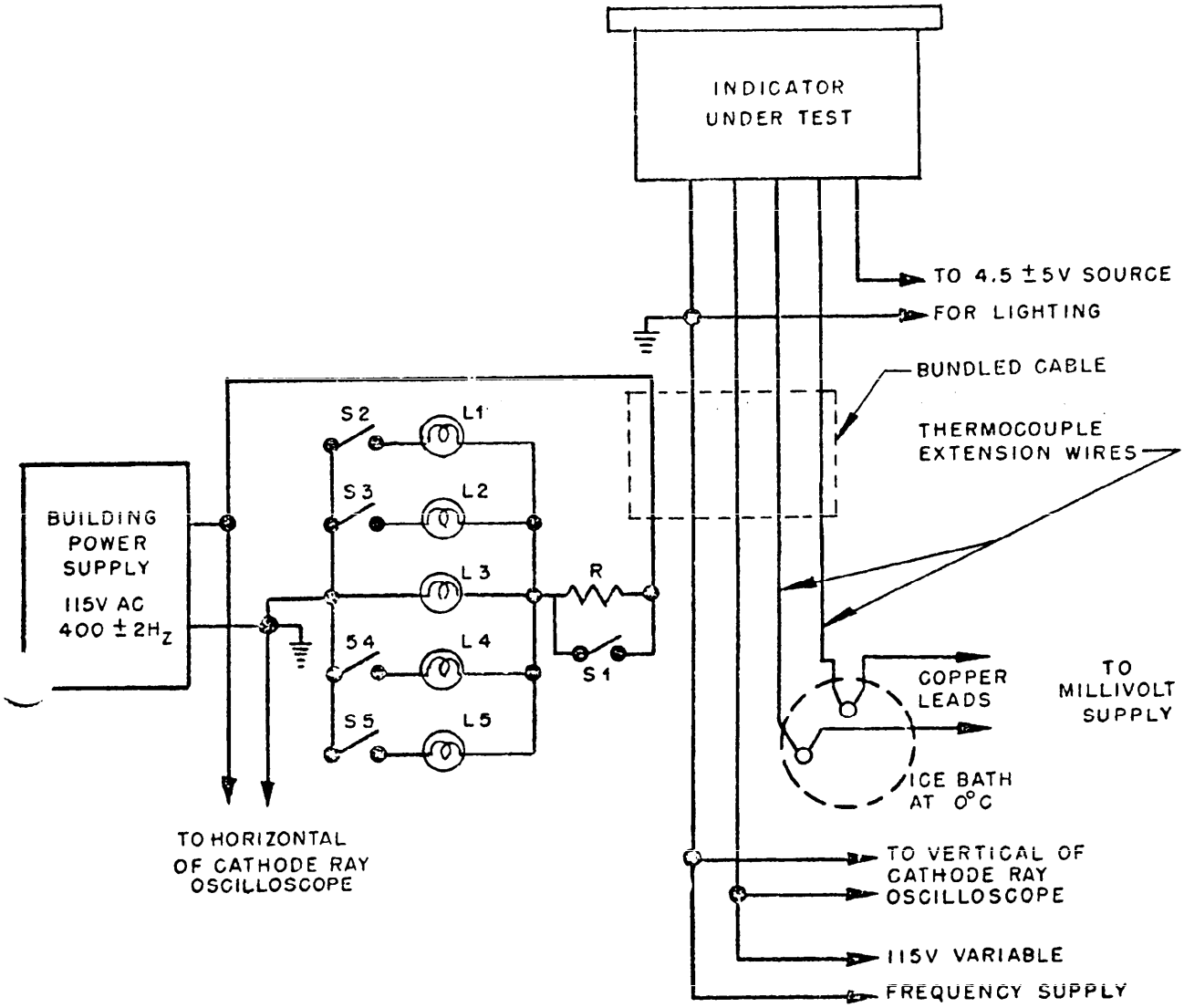
.6.25 Frequency cycling, AC indicators. The voltage shall be adjusted to 115 ±1V and the frequency shall be cycled from 320 to 480 Hz in accordance with figure 2 with an indication of 750°C. One cycle shall consist of frequency variations from time zero to 100 seconds and return to time zero through the time-frequency curve of figure 2. This procedure shall be repeated at a rate of 18 cycles per hour for 8 hours. Power shall then be removed for a minimum period of 2 hours and shall be reapplied at 121 ±1V, 320 ±5 Hz for 1 hour. Immediately following this 1-hour period, the indicator shall be frequency cycled for 1 hour as indicated herein with 115 ±1V applied. Any damage which would prevent compliance with the scale-error tolerances for room temperature shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.26 Voltage surge, AC indicators. Power of 108 ±1V, 480 ±5 Hz shall be applied for 1 hour followed immediately by application of 121 ±1V, 480 ±5 Hz for an additional hour. The indication shall be 750°C during this test. At the end of the second 1-hour period, the voltage and frequency shall be adjusted to 115 ±1V, 320 ±5 Hz. A voltage surge shall be applied to the power input to provide a momentary root mean square (rms) voltage of 160 ±5V, and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.09 ±0.005 second. The frequency shall then be increased to 430 ±5 Hz and a voltage surge shall be applied to provide a momentary rms voltage of 195 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.07 ±0.005 second. At the end of this test, the voltage and frequency shall be adjusted to 115 ±1V, 400 ±5 Hz. The scale error test shall be conducted at room temperature. The scale errors shall not exceed those specified in the detail specification.

4.6.27 Stray signal and beat frequency toleration, AC indicators. The indicator shall be connected as shown on figure 1. All switches shall be open. With the building power supply turned off, 115V 400-cycle power and a simulated thermocouple signal that will produce an indication of 750°C shall be applied to the indicator. After 2 minutes at the 750°C indication, the cathode ray oscilloscope and the building power supply shall be turned on and allowed to warm up for an additional 2 minutes. Beginning with a frequency of 380 Hz, the frequency of the variable frequency supply shall be slowly increased at a maximum of 1 Hz every 5 seconds until 420 Hz is reached. The temperature indication shall be closely observed throughout the test. Any change in indicator pointer position, oscillation of the pointer, or sluggishness shall be recorded and the value of frequency shall be noted. This procedure shall be repeated with only S-1 closed; S-1 and S-2 closed; S-1, S-2, and S-3 closed; S-1, S-2, S-3, and S-4 closed; and then with all switches closed. During this test, each of the thermocouple leads shall be grounded alternately with each group of switches closed. Pointer or counter oscillation shall be cause for rejection. The maximum change in indication throughout the test shall not exceed 4°C. If sluggishness is observed during any of the specified conditions, the response time test shall be conducted with the worst condition maintained. The response time shall not exceed 2.5 seconds. With only the S-1, S-2, and S-3 switches closed, a section of the cable containing all of the conductors except the 5V lighting input conductor shall be laid against the entire length of the indicator and then across the indicator at various positions on the case length. Pointer or counter oscillation and variation shall be observed at each position of the cable as the frequency is varied from 380 to 420 Hz as specified herein. Pointer or counter oscillation shall not exceed 5°C double amplitude and pointer or counter variation shall not exceed 3°C. The test specified in 4.6.4 shall then be conducted with the worst condition maintained and the response time shall not exceed 2.5 seconds. The 400 ±2 Hz power supply shall be turned off and the conductor cable shall be extended to the rear of the indicator.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.



NOTES:

BUNDLED CABLE - CONDUCTORS LAID PARALLEL AND CENTRALLY ARRANGED IN TIGHTLY BUNDLED CABLE FOR 20 FT. MINIMUM MIL-W-5846, TYPE II, CLASS A.

R = 100,000 ±5 000 OHM
L1, L2, L4, & L5 = 300 WATT LAMPS OR EQUIVALENT IMPEDANCES
L3 = 60 WATT LAMP OR EQUIVALENT IMPEDANCE

FIGURE 1. Stray signal and beat frequency test wiring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

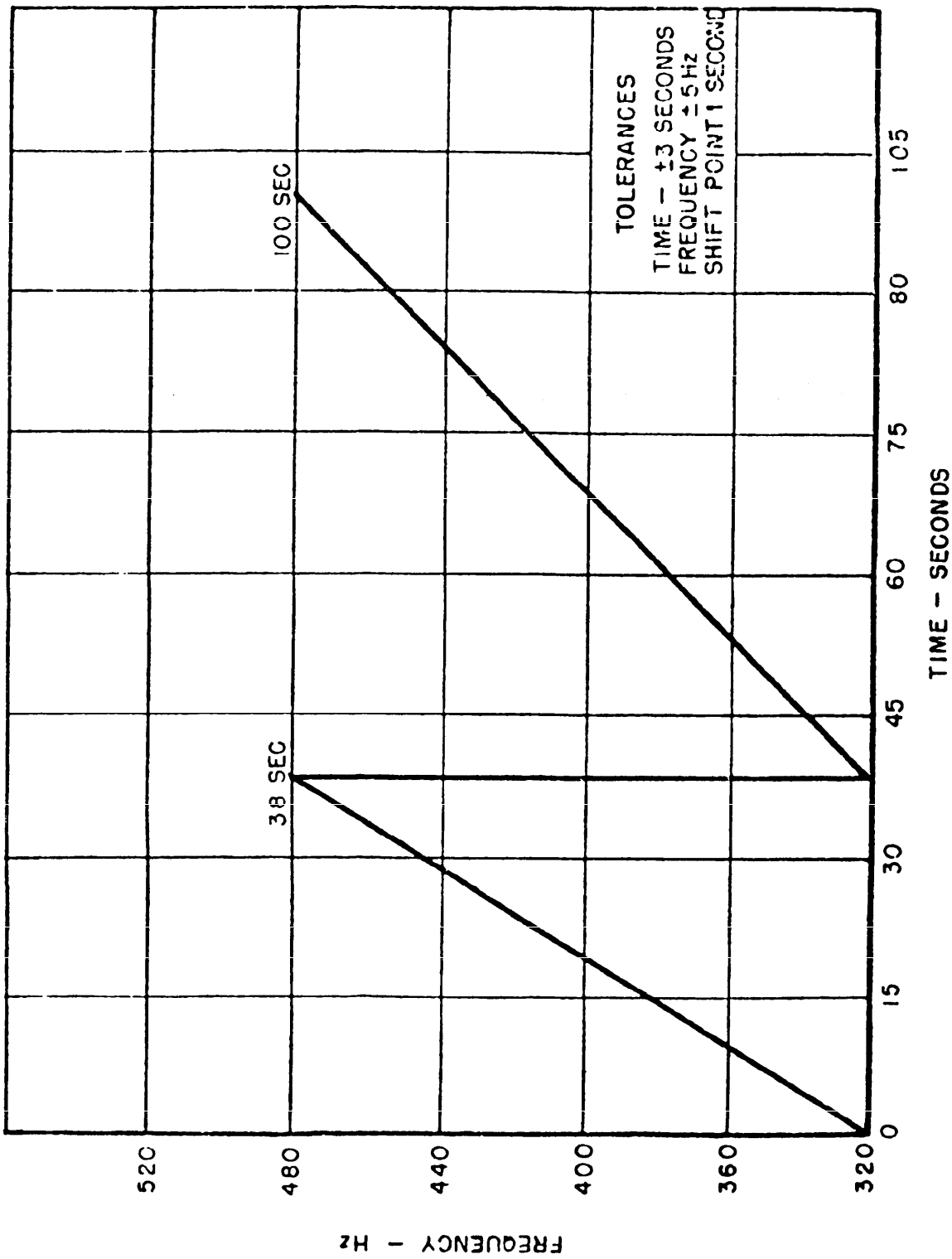


FIGURE 2. Time-Frequency curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

The integral lighting circuit shall then be energized with 4.5 ±0.5 at 400 ±2 Hz while the indicator power supply frequency is varied from 380 to 420 Hz. Pointer or counter oscillation shall be cause for rejection. Pointer or counter variation shall not exceed 1°C.

4.6.28 Thermal shock. The indicator shall be subjected to eight cycles of exposure to water at 85°C or greater and 5°C or less without evidence of moisture penetration or damage to coating or enclosure. Each cycle of the test shall consist of immersing the indicator in water at 85°C for 30 minutes and then within 5 seconds of removal from the bath, immersing it 30 minutes in the other bath maintained at 5°C. This cycle shall be repeated continuously, one cycle following the other until eight cycles have been completed. Following this test, the indicator shall be subjected to the sealing test. No leakage shall be detected nor shall there be any damage to the indicator as a result of this test.

4.6.29 Fogging. The indicator shall be placed in a 71° ±2°C controlled ambient temperature for a minimum of 1 hour. After this period, and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of 1 to 2 minutes. The glass shall be wiped dry immediately (do not use compressed air) and the indicator inspected for evidence of water or oil fog or condensate inside the cover glass. Evidence of fogging or condensate shall be cause for rejection.

4.6.30 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, procedure I for a period of 48 hours. At the conclusion of this test, the test specified in 4.6.4 shall be conducted. Tolerances shall not exceed those specified in the detail specification for scale error at room temperature.

4.6.31 Humidity test. Indicators which have passed the individual tests shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.1, procedure I. Within 1 hour after the test, the indicator shall be subjected to the tests specified in 4.6.4. There shall be no evidence of moisture on the inside of the cover glass.

4.6.32 Fungus test. The indicator shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508.1, procedure I for a period of 14 days. The indicator shall not support fungus growth. Written vendor certification that all materials used are non-nutrient will be considered toward waiver of this test.

4.6.33 Reliability. Eight indicators shall be tested for reliability. Acceptance permitting production and the total test time shall be determined in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.34 Test procedure. The test equipment and fixtures shall include provisions for vibration and temperature cycling at periodic intervals as specified herein. For the vibration portion of this test, a small instrument panel and a motor with an asymmetric weight that will produce the required amplitude and frequency as measured at the indicators shall be mounted to a strong flat plate which is supported on vibration isolators. The double amplitude shall be 0.010 to 0.015 inch with a frequency of 45 to 55 Hz. The amplitude and frequency shall be such as to provide a vibration level of 1.5g ±10 percent. The direction of vibration is not critical. The indicators shall first be subjected to the scale-error test at room temperature and then to operational cycles consisting of 5.5 hours of operation with power and millivoltage applied and 30 minutes with power off. The operational cycles shall be conducted at 0°C, room temperature, and 50°C in the following sequence: 5.5 hours at 0°C, 30 minutes off; 5.5 hours at room temperature, 30 minutes off; 5.5 hours at 50°C, 30 minutes off; and 5.5 hours at room temperature, 30 minutes off. During the operating portion of this cycle, the millivoltage shall be varied to produce an indication cycle corresponding to a change ranging from 200°C to 1,000°C and return to 200°C at a rate of 2.5 ±0.5 cpm. This shall constitute one cycle. The indication and operational cycles shall be repeated continuously until an accept or reject decision is reached. The continuity of this test may be interrupted for weekend shutdown. During each 5.5-hour operational cycle, each indicator shall be vibrated for a minimum period of 10 minutes each hour. During the 30-minute-off periods, the indicators shall be subjected individually to the scale error test at room temperature so that all indicators will be tested once each 100 hours. The scale errors recorded throughout this test shall not exceed the tolerances specified in the detail specification for scale error during reliability testing. Upon completion of this test, the indicators shall meet the tests specified in 4.6.4 and 4.6.7. Malfunction or failure of any component or part of the indicator during this test shall be counted as a relevant failure, except that a lamp failure in the dial lighting circuit after 1,000 hours shall not be counted as a failure.

4.6.35 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471. The procedure shall be selected by the contractor and submitted to the procuring activity for approval.

4.6.36 Lighting (brightness and color)

4.6.36.1 The following tests shall be performed as individual tests during qualification and sampling plan A, B, and C tests:

4.6.36.1.1 Light. The indicator shall be subjected to the light tests (brightness, color, and stray light) as specified in MIL-L-27160 and MIL-L-25467 as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.36.1.2 Subarea measurements. All areas including the indicator areas specified in 3.14 shall be measured.

4.6.36.2 The following tests shall be conducted only during the individual production quality conformance testing:

4.6.36.2.1 Visual comparison. This test shall be conducted by qualified personnel only after they have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark adapted. The prime standard and the indicator under test shall be placed side-by-side at eye level in completely dark surroundings and viewed at a distance of approximately 2.5 feet in a sighting cone 45° from the normal to the surface of the cover glass. All indicator markings, pointers, flags, et cetera, shall be compared when viewed from any position within the sighting cone.

4.6.36.2.2 Brightness. With 2.50 ±0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the standard, like areas of the test indicator and standard shall be compared such as: pointers, command bars, alarm flags, numerals, indicia, markings, et cetera for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness levels of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar or exceeds the limits of the standard in any individual area or the relative uniformity of lighting of the indicator and prime standard are different. The questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this test, they shall be rejected and returned for rework. Qualification units shall not be tested by use of the prime standard. The qualification units must be photometrically measured in accordance with the applicable lighting specification.

4.6.36.2.3 Color. The color comparison test shall be conducted as specified in 4.6.36.1.1, except 4.50V shall be applied to both the production indicator and the prime standard. The two indicators shall appear identical in color.

4.6.37 Detail examination. The indicator shall be critically examined to determine full compliance with regard to weight, physical dimensions, dial composition, and similar detail requirements. The indicator shall be examined internally for evidence of corrosion or deterioration resulting from the tests specified herein. There shall be no evidence of corrosion or deterioration.

4.7 Preparation for delivery. Preservation, packaging, packing, and marking shall be examined for conformance to Section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified by the procuring activity (see 6.2).

5.1.1 Level A. Each indicator shall be packaged in accordance with method III of MIL-P-116. Each unit, appropriately cleaned, shall be placed within a loosely fitting, transparent plastic film bag, material conforming to type III of MIL-F-22191, minimum thickness 0.003 inch. The air within the bag shall be exhausted to the extent practical prior to accomplishing the bag closure. Each indicator shall be wrapped within a minimum of 1-inch thickness of cellulose cushioning material conforming to PPP-C-843 and then placed within an individual snug-fitting carton conforming to PPP-B-636, type CF, class domestic. Alternate cushioning material of equivalent protective quality may be substituted. The continuous line marking requirement of MIL-F-22191 is not mandatory.

5.1.2 Level C. Each indicator shall be packaged in a manner that will afford the item the degree of protection required to prevent chemical corrosion and physical deterioration during shipment from the supply source to the first receiving activity. The supplier's commercial practice may be utilized when it conforms to the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified by the procuring activity (see 6.2).

5.2.1 Level A. Twenty-four or 36 indicators, packaged as specified in 5.1.1, shall be packed for shipment in containers conforming to type CF or SF, weather-resistant class V3s of PPP-B-636. The weight and dimension limitation criteria of table II shall apply. When specified (see 6.2), indicators shall be packed in containers conforming to PPP-B-601, overseas type. Insofar as practicable, exterior containers shall be of uniform shape and size and of minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed for shipment in containers conforming to PPP-B-636 as indicated in 5.2.1, except that the shipping containers shall be domestic service grade, domestic class. Dimension and weight limitation criteria shall be as stated in table I of this specification. The sealing requirements shall not apply.

5.2.3 Level C. Indicators packaged as specified shall be packed for shipment in a manner that will afford adequate protection during shipment from the supply source to the first receiving activity. This level shall conform to the applicable carrier rules and regulations. The contractors commercial practice may be utilized when such meets the requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as specified in the detail specification.

5.3.1 In addition to the standard marking, all containers shall be marked with the following cautionary warning:

GLASS - FRAGILE INSTRUMENT - HANDLE WITH CARE

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use on aircraft to indicate various temperature functions, such as engine bearing and exhaust temperatures.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. When sampling plan B tests are to be conducted (see 4.5.2.2)

c. Levels of packaging and packing required (see section 5). PPP-B-601 boxes should be used only when specified in the contract or order (see 5.2.1)

d. When sampling plan A or sampling plan C tests shall be conducted.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASD/ENAID, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer activity:
Air Force - 99

Preparing activity:
Air Force - 11

Project No. 6685-F502

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.